

RESEARCH ARTICLE

Hangeshashinto (TJ-14) prevents radiation-induced mucositis by suppressing cyclooxygenase-2 expression and chemotaxis of inflammatory cells

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Abstract

Purpose Radiation-induced oral mucositis is the most common side effect of radiotherapy in head and neck cancer; however, effective modalities for its prevention have not been established. In this study, we evaluated the effectiveness of Hangeshashinto (TJ-14), a Japanese herbal medicine, for preventing radiation-induced mucositis and elucidated its effect on inflammatory responses, including inflammatory cell chemotaxis and cyclooxygenase-2 (COX2) expression, in an animal model.

Methods Syrian hamsters, 8–9 weeks old, were enrolled in this study. Animals were irradiated with a single 40 Gy dose to the buccal mucosa. Hamsters freely received a treatment diet mixed with 2% TJ-14 or a normal diet daily. The therapeutic effect was determined based on the visual mucositis score, body weight, and histological examination of infiltrated neutrophils and COX2 expression.

Results TJ-14 significantly reduced the severity of mucositis. The percentage with severe mucositis (score ≥ 3) was 100% in the untreated group and 16.7% in the TJ-14 group ($P < 0.05$). There was no difference in body weight change between the groups; however, weight gain in the untreated group tended to be suppressed compared to that in the TJ-14 group during the peak period of mucositis.

In addition, TJ-14 inhibited the infiltration of neutrophils and COX2 expression in irradiated mucosa ($P < 0.05$).

Conclusions TJ-14 reduced the severity of mucositis in an animal model by suppressing the inflammatory response. Because TJ-14 is inexpensive and its safety is established, it is a promising candidate for the standard treatment of radiation-induced mucositis in cancer patients.

Keywords Radiotherapy · Radiation-induced mucositis · Kampo · Supportive care · Cancer treatment · Inflammation

Introduction

Radiation-induced oral mucositis is a side effect that affects virtually all patients who receive radiotherapy for head and neck cancer (HNC). During the past two decades, chemoradiotherapy (CRT) has contributed to an improvement in local control and overall survival in the advanced stages of HNC [1, 2]; however, the rate of severe mucositis has increased to 70% in patients receiving CRT, whereas the rate is 40% in those receiving only radiotherapy [3]. Although it is well known that an aggressive regimen of chemotherapy, such as that used for hematopoietic stem-cell transplant, also causes severe mucositis, patients receiving radiotherapy for HNC tend to be affected for a longer period because the mucosal damage by fractionated irradiation accumulates during the treatment period and lasts up to 3 weeks after cessation of treatment [4]. Radiation-induced mucositis in HNC patients is unavoidable because of the proximity of the oral cavity, pharynx, and larynx to the cancer lesion. Severe pain and dysphasia often affect the patient's quality of life. Furthermore, malnutrition and infection may result in the interruption of or unplanned changes to the treatment schedule, which may

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lead to an undesired influence on local control of the cancer [5]. Despite these clinical problems associated with mucositis, an efficient method for preventing mucositis has not been established. According to the updated guidelines for oral mucositis from the Multinational Association of Supportive Care in Cancer and the International Society of Oral Oncology (MASCC/ISOO) [6], several supportive care options, including benzydamine mouthwash, oral care protocols, and morphine mouthwash, are recommended for patients with HNC. However, these interventions are limited in their analgesic effectiveness and the clinical benefits are controversial [4]. Recombinant human keratinocyte growth factor-1 (rKGF-1/palifermin) has been approved to repair radiation-induced mucosal damage and is also recommended for chemotherapy-induced mucositis. However, its application for epithelial malignancies, including HNC, was not mentioned in the guidelines because of a potential concern for adverse effects on the tumor tissue [7]. Moreover, the high cost of rKGF-1 may offset the economic benefit of preventing mucositis.

Hangeshashinto (TJ-14), a traditional Japanese herbal medicine (*Kampo* medicine), is composed of seven medicinal herbs: *Pinelliae tuber*, *Scutellariae radix*, *Zingiberis processum rhizoma*, *Glycyrrhizae radix*, *Zizyphi fructus*, *Ginseng radix*, and *Coptidis rhizoma*. It is commonly prescribed for patients with inflammatory disorders of the digestive tract, such as stomatitis, gastritis, and diarrhea, and is covered by public health insurance in Japan. Recently, the therapeutic benefit of TJ-14 for chemotherapy and radiotherapy-induced mucositis has been reported [8, 9] and it may be a promising candidate for mucositis treatment in cancer patients; however, its mechanism of action has not been determined.

In this study, we investigated the preventive effect of TJ-14 on radiation-induced mucositis in an animal model and evaluated its effect on the inflammatory response, including cyclooxygenase-2 (COX2) and inflammatory cell chemotaxis, in mucosal tissue.

Materials and methods

Animals

The study protocols were reviewed and approved by the Animal Ethical Committee of the National Defense Medical College (No. 13075) and were carried out in accordance with the Guidelines for Proper Conduct of Animal Experiments and the US Public Health Service Policy on Human Care and Use of Laboratory Animals. Female 8- to 9-week-old Syrian golden hamsters (SRL, Tokyo, Japan) were habituated for 3 days before the study and housed two animals per cage. The animals were maintained in a 12 h

light/dark scheduled and humidity-controlled environment at a temperature of 21 ± 1 °C.

Irradiation to the buccal mucosa

Animals were anesthetized intraperitoneally with ketamine (75 mg/kg) and medetomidine (1.0 mg/kg) and placed in a lateral position. The left buccal pouch was everted, extended manually, and stabilized with a rubber plate. The remainder of the body was shielded from radiation by a lead plate. A single dose (40 Gy) of radiation was delivered to the buccal mucosa using an X-ray apparatus (MBR-1505 R2; Hitachi Medical, Tokyo, Japan) at a dose rate of 1.0 Gy/min. The tube voltage and current were 150 kV and 5 mA, respectively, and the X-ray was filtered using 2.0 mm aluminum.

Reagents

TJ-14 was provided by Tsumura & Co. (Tokyo, Japan). It is a mixture of the following seven medicinal herbs: *P. tuber*, 27%; *S. radix*, 13.5%; *Z. processum rhizoma*, 13.5%; *Glycyrrhizae radix*, 13.5%; *Z. fructus*, 13.5%; *Ginseng radix*, 13.5%; and *C. rhizoma*, 5.4%. The treatment diet was a mixture of 2% TJ-14 in the normal diet (CR-LPF; Oriental Yeast Co., Ltd; Tokyo, Japan) and was shaped into a pellet identical to that of the normal diet.

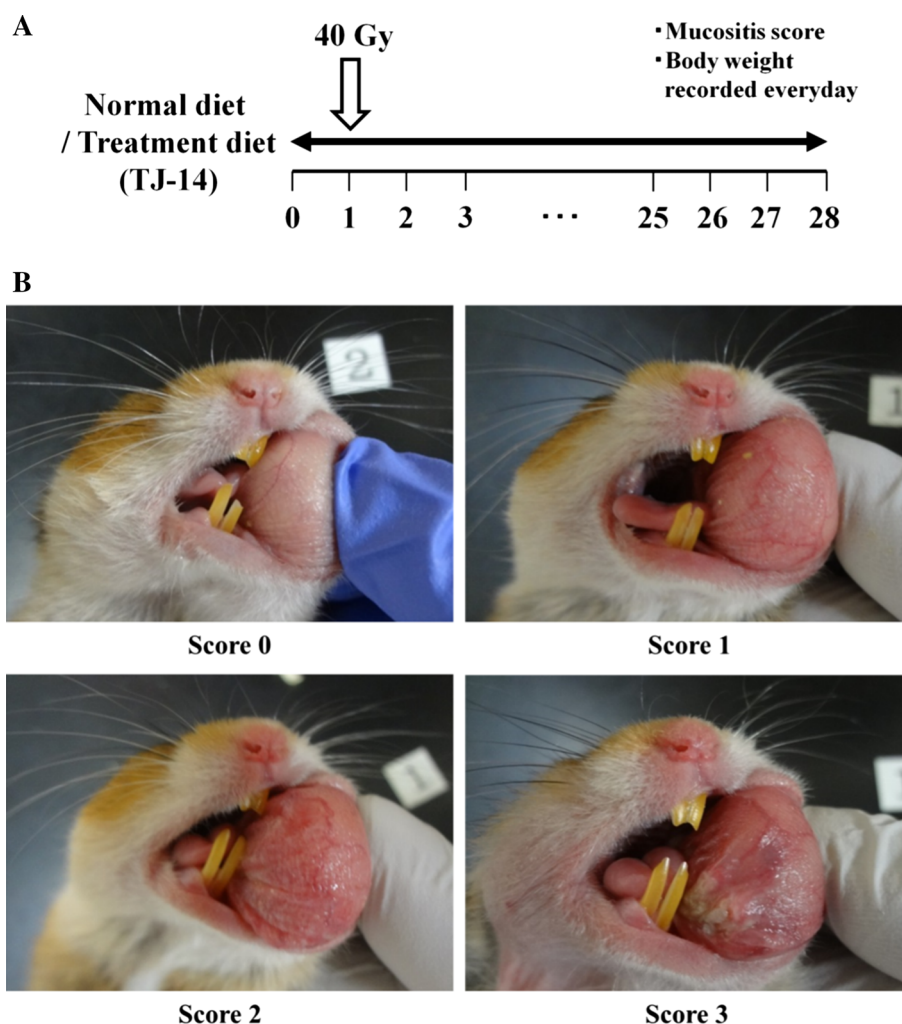
Protocol of the in vivo experiment

The protocol of the in vivo experiment is shown in Fig. 1a. The animals were divided randomly into two groups. The untreated group ($n = 6$) was administered a normal diet and the TJ-14 group ($n = 6$) was administered the treatment diet (2% TJ-14). Both diets were administered from day 0 to day 28 and animals were allowed to eat freely during the study. A single 40 Gy dose of radiation was delivered to the left buccal mucosa on day 1 under general anesthesia. Body weight and photographs of the buccal mucosa were recorded every day through day 28. After the study, two blinded independent observers examined and scored the visual appearance of the mucosa in each randomized photograph.

Visual scoring of the irradiated mucosa

Visual examination of the irradiated buccal mucosa was carried out once a day. Animals were anesthetized with isoflurane inhalation and the buccal mucosa was everted and examined (Fig. 1b). Digital images were always taken in the same light environment and the severity of the mucositis was scored using an established visual scale (Fig. 2a) [10, 11]. The scores ranged from 0 to 4 as

Fig. 1 In vivo study of radiation-induced mucositis using a hamster model.
a Experimental protocol. Normal diet or treatment diet (TJ-14) was administered from day 0 and a single dose of radiation was given on day 1.
b Representative macroscopic appearance and scores of radiation-induced mucositis in buccal mucosa



follows: 0, completely healthy mucosa; (1) light to severe erythema and vasodilation, no erosion of the mucosa; (2) severe erythema and vasodilation, erosion of superficial aspects of the mucosa leaving denuded areas and decreased stippling of the mucosa; (3) severe erythema and vasodilation, including formation of off-white ulcers, which may be pseudomembranous, in a cumulative area of about one quarter of the cheek pouch; (4) severe erythema and vasodilation with ulcers covering about one half of the cheek pouch, loss of mucosal pliability.

Histological observation

Tissue specimens were obtained from the buccal mucosa of three groups of animals: a normal group (unirradiated mucosa), an untreated group (irradiated TJ-14 untreated mucosa), and a TJ-14 treated group (irradiated TJ-14 treated mucosa). Irradiated mucosa in the untreated group and the TJ-14 treated group was excised 16 days after 40 Gy irradiation (the day of

maximum severity of mucositis as indicated by the pilot study). Tissue samples were histologically evaluated from a total of four animals from each group. The excised tissues were fixed in 4% paraformaldehyde, dehydrated by ethanol and xylene, and embedded in paraffin block. Serial sections (5 μ m thick) were obtained for hematoxylin–eosin staining (HE), naphthol AS-D chloroacetate esterase (NASD) staining, and immunohistochemistry of COX2.

NASD staining and quantification of inflammatory cells

To evaluate the infiltration of neutrophils, tissue was stained with NASD using a commercially available staining kit (Muto Pure Chemical, Co., Ltd., Tokyo, Japan) and the sections were counter-stained using Mayer's hematoxylin solution. Quantitative analysis of infiltrating neutrophils was also performed. For each specimen, 5 square areas (200 $\times$ 200 μ m) were randomly selected from the

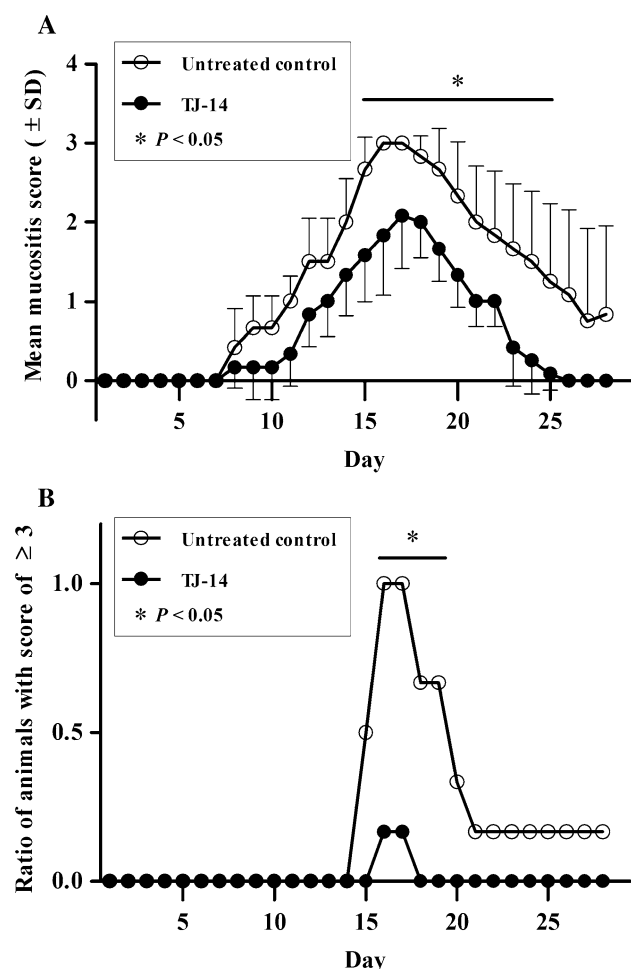


Fig. 2 Effect of TJ-14 on severity of radiation-induced mucositis. **a** Mean mucositis scores. **b** Percentage of animals with a score ≥ 3 . Animals were given the treatment diet (TJ-14) in the TJ-14 group and the normal diet in the untreated control group. * $P < 0.05$, TJ-14 treatment vs untreated control. Wilcoxon rank-sum test for mean scores and Fisher's exact test for ratio of severe mucositis were used for statistical analysis. Error bars represent SD

submucosa within a 500- μ m-depth from where mucosa was denuded or ulcerated. The acquired images were processed and quantified using the National Institutes of Health (NIH) ImageJ software (NIH, Bethesda, MD, USA).

Immunohistochemistry of COX2

To evaluate COX2 expression in tissue, immunohistochemical staining was performed using a Vectastain Elite ABC Rabbit IgG Kit (PK-6101; Vector Laboratories, Inc., Burlingame, CA, USA). Sections of tissue (5 μ m thick) were deparaffinized, hydrated, and autoclaved at 120°C for 5 min and then incubated with 3% hydrogen peroxide in methanol for 5 min to block endogenous peroxidase. Goat serum was used to block nonspecific binding. The sections

were incubated for 2 h with anti-COX2 polyclonal rabbit IgG diluted 1:400 (ab15191; Abcam, Cambridge, MA, USA). Secondary antibody (biotinylated anti-rabbit IgG) was added and the sections were further incubated for 30 min at 23°C. Thereafter, the sections were treated with avidin and biotinylated horseradish peroxidase complex for 30 min, then stained using 3,3-diaminobenzidine (ImPACT DAB; Vector Laboratories, Inc.) for 1 min, counterstained with Mayer's hematoxylin, and observed using a light microscope.

Quantification of COX-2 expression by western blot analysis

Buccal mucosa was resected from animals in the normal group (unirradiated mucosa), untreated group (irradiated TJ-14 untreated mucosa), and TJ-14 group (irradiated TJ-14 treated mucosa) ($n = 4$) on day 16 and stored at -80°C . Tissues were crushed in liquid nitrogen and homogenized in high salt lysis buffer. Supernatant was collected and protein concentrations were determined colorimetrically using a BCA Protein Assay Kit (Thermo Fisher Scientific, San Jose, CA, USA). Samples were processed at 70 °C for 10 min with NuPAGE LDS sample buffer (Invitrogen, Carlsbad, CA, USA). The samples were applied (50 μ g per lane) to 10% sodium dodecyl sulfate gels and transferred to polyvinylidene difluoride membranes. Membranes were blocked in tris-buffered saline-Tween (TBS-T) with 5% skimmed milk and incubated with a 1:1000 dilution of primary antibodies, anti-COX2 polyclonal rabbit IgG (ab15191; Abcam, Cambridge, MA, USA) or anti- β -actin rabbit IgG (#4967; Cell Signaling Technology, Beverly, MA, USA), for 1 h at 23°C. After washing, membranes were incubated with secondary antibody, anti-rabbit IgG HRP-linked antibody (#7074S; Cell Signaling Technology), for 1 h at 23°C. Protein bands were detected using Super Signal West Dura (Pierce, Rockford, IL, USA) chemiluminescent reagent and analyzed using a LAS4000 mini digital imaging system (Fujifilm, Tokyo, Japan).

Statistical analysis

All statistical analyses were carried out using JMP ver.10 software (SAS Institute, Cary, NC, USA). Statistical significance between the groups was analyzed using the Wilcoxon rank sum test for mucositis score, Fisher's exact test for ratio of severe mucositis, and Student's t tests for body weight, neutrophil count, and COX2 expression. P values less than 0.05 were considered statistically significant.

Results

Therapeutic effect of TJ-14 on radiation-induced mucositis in vivo

In this animal model, irradiated mucosa reliably developed mucositis as described in previous reports [10, 11]. Erythema and vasodilation were observed from day 8 and peak levels of mucositis with ulcerative lesions developed between days 15 and 20. Mucositis healed 4 weeks or more after irradiation. TJ-14 was administered by a mixed treatment diet to the animals in the TJ-14 group starting 1 day before irradiation (day 0). As shown in Fig. 2a, mean mucositis scores in the TJ-14 group were lower than those in the untreated group and a statistically significant difference was noted from day 15 to 25 ($P < 0.05$). The percentage with severe mucositis with ulcerative lesions (score ≥ 3), which corresponded to CTCAE v3.0 grade ≥ 3 , was 100% in the untreated group and 16.7% in the TJ-14 treated group at day 16. A statistically significant difference in the percentage with severe mucositis was observed from day 15 to day 18 (Fig. 2b).

Body weight change after irradiation

Body weight changes from day 0 to 28 are shown in Fig. 3. No statistically significant difference was observed between the groups; however, the untreated group showed a slight tendency toward suppressed weight gain around day 15, when the severity of mucositis reached its peak. In contrast, the animals in the TJ-14 treated group stably gained weight regardless of the peak severity of mucositis.

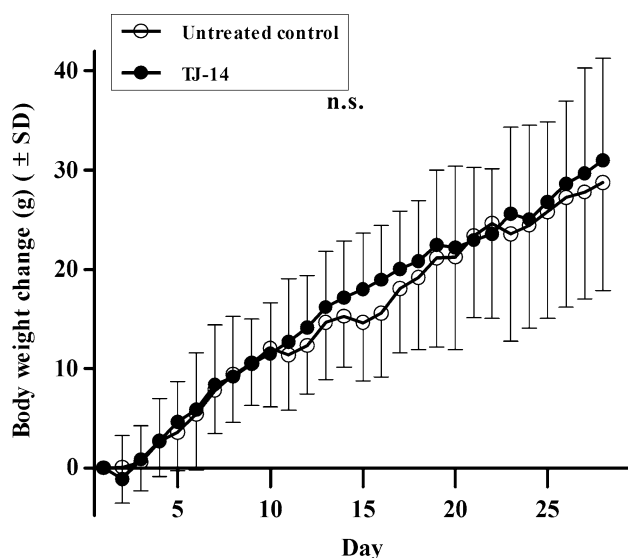


Fig. 3 Body weight change after irradiation. Although no statistically significant difference was observed, the TJ-14 group stably gained weight regardless of the presence of mucositis. Error bars represent SD

Evaluation of neutrophil infiltration in the irradiated mucosa

To histopathologically evaluate the normal mucosa (unirradiated mucosa), untreated mucosa (irradiated TJ-14 untreated mucosa), and TJ-14 treated mucosa (irradiated TJ-14 treated mucosa), buccal mucosal tissues were excised and analyzed on day 16. Irradiated mucosal tissue exhibited an enucleated epithelium and ulceration with a large number of infiltrated inflammatory cells in the submucosa (Fig. 4a, upper row). NASD staining indicated a sparse distribution of neutrophils in normal non-irradiated mucosa; however, robust infiltration was observed in the epithelial and submucosal tissue from irradiated mucosa. Notably, neutrophil infiltration was significantly reduced in TJ-14 treated mucosa compared to untreated mucosa ($P < 0.05$) (Fig. 4a, middle row). Mean neutrophil count (cells/ $4 \times 10^4 \mu\text{m}^2$) was 3.0 ± 1.0 (SD) in normal mucosa, 142.6 ± 33.5 (SD) in untreated mucosa, and 63.6 ± 37.3 (SD) in TJ-14 treated mucosa (Fig. 4b).

Evaluation of COX2 expression in irradiated mucosa

COX2 expression in normal, untreated, and TJ-14 treated mucosa at day 16 was assessed by immunohistochemical staining and western blot analysis. As shown in Fig. 5a (lower row), robust expression of COX2 was observed, especially in migrated fibroblasts located in the submucosa of untreated irradiated mucosa, whereas no evident expression was noted in normal mucosa. However, COX2 expression in irradiated mucosa was reduced in the TJ-14 treated group compared to the untreated group. Western blot analysis indicated that the relative the expression of COX2 was 1.0 ± 0.32 (SD) in the untreated group and 0.34 ± 0.18 (SD) in the TJ-14 group ($P < 0.05$) (Fig. 5a, b).

Discussion

In this study, we found that oral administration of TJ-14 prevented radiation-induced mucositis in an animal model. In addition, a significant reduction in COX2 expression and neutrophil infiltration was observed. These results suggested that the anti-inflammatory property of TJ-14 contributed to the prevention of mucositis.

Many supportive care options have been reported for radiation-induced oral mucositis; however, definitive treatment has not established [12] despite the impact on the patient's quality of life, medical costs, and outcome of cancer treatment. The pathogenesis of oral mucositis associated with cancer therapy is a complex biological

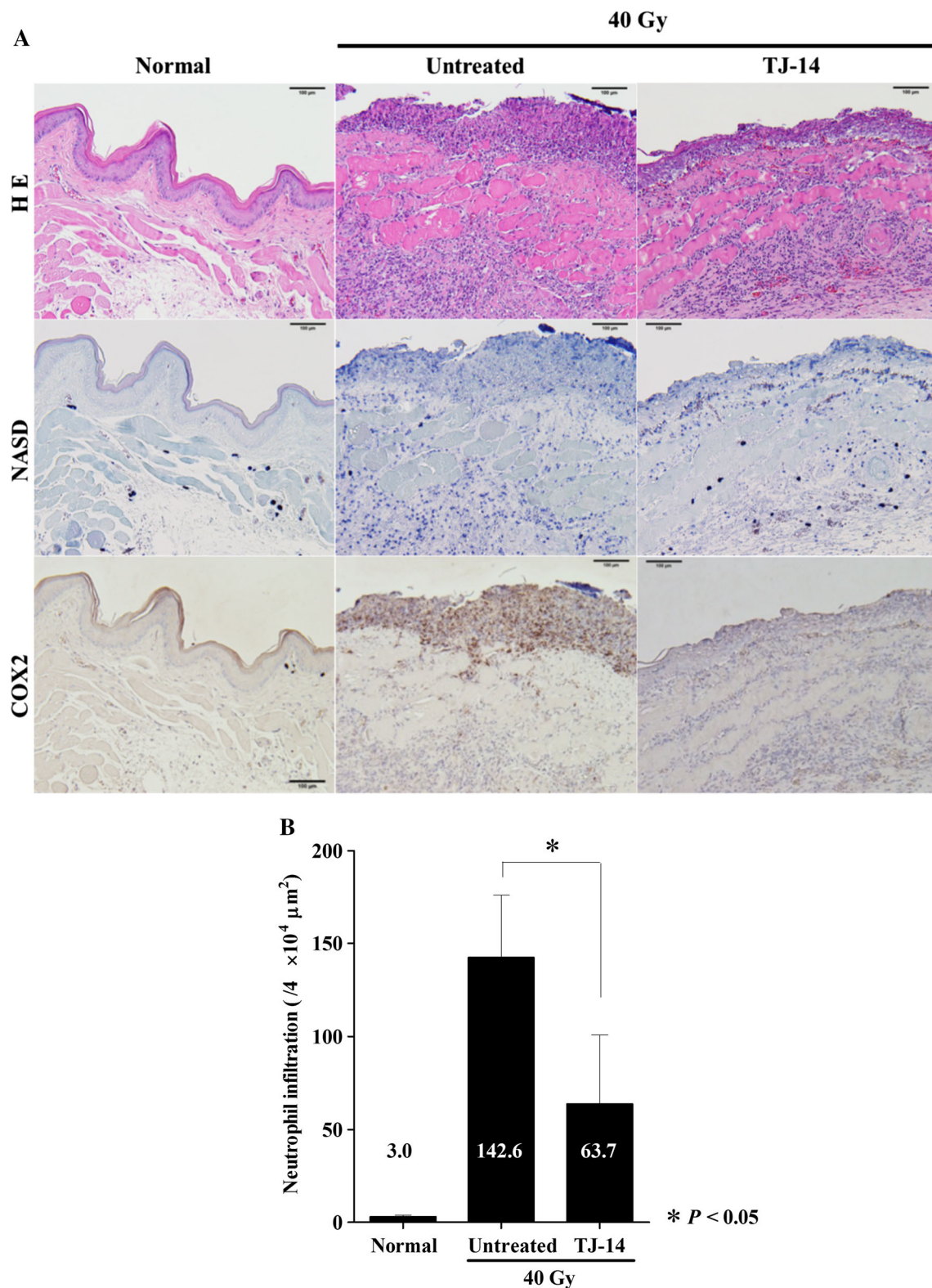
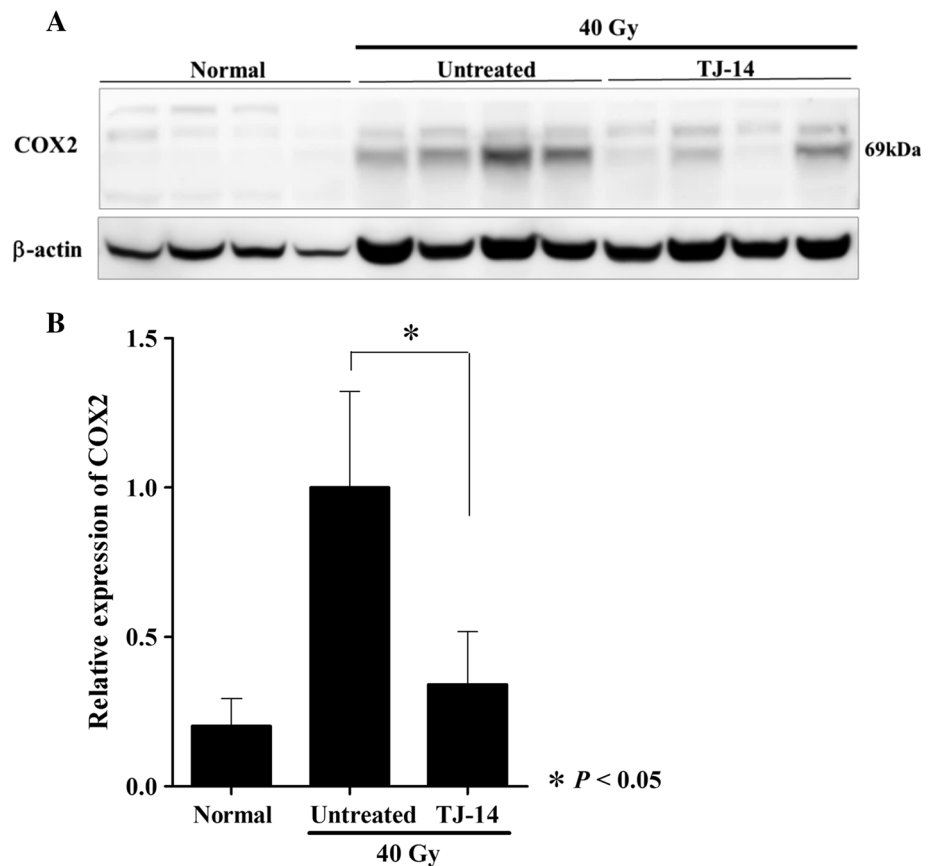


Fig. 4 Histopathological assessment. **a** Normal vs irradiated buccal mucosa at day 16. Serial HE sections are shown in the *upper row*, NASD sections for visualizing the infiltrated neutrophils are shown in the *middle row*, and immunohistochemical analysis of COX2 expression is shown in the *lower row*. Scale bar represents 100 μm .

b Mean counts of infiltrated neutrophils in normal mucosa, untreated mucosa, and TJ-14 treated mucosa. A significant decrease in infiltrated neutrophils was observed in TJ-14 treated mucosa after irradiation. * $P < 0.05$, TJ-14 treated vs untreated mucosa assessed by Student's *t* test. Errors bars represent SD

Fig. 5 COX2 expression in buccal mucosa was assessed for normal and irradiated buccal mucosa on day 16. **a** Western blot gel image of COX2 expression in normal, TJ-14 treated, and untreated mucosa. **b** Quantification of COX2 levels. COX2 expression was suppressed in irradiated TJ-14 treated mucosa. * $P < 0.05$, TJ-14 treated vs untreated mucosa by Student's t test. Error bars represent SD



process as reviewed by Sonis et al. [13]. Briefly, the first step is triggered by reactive oxygen species (ROS) and free radicals induced by radiation and chemotherapy. The ROS and free radicals damage the DNA strands in the rapidly proliferating cells of the epithelium. This oxidative stress upregulates and activates early response genes, transcriptional factors, inflammatory cytokines, and signal transduction pathways, including the COX, mitogen-activated protein kinases (MAPKs), nuclear factor- κ B (NF κ B), and ceramide synthase pathways. This results in a disruption to the mucosal epithelium. Bacterial infection and infiltrated inflammatory cells, such as neutrophils and macrophages, amplify the inflammatory response in a positive feedback loop, which, in turn, causes more severe ulcerative lesions and delays wound healing. Therefore, because the development of mucositis is a complex biological process that involves many signals, the ideal pharmacological approach is multi-targeted.

As indicated in this study, COX2 was strongly expressed in irradiated mucosa and was significantly inhibited in TJ-14 treated mucosa compared to untreated irradiated mucosa. COX2 causes the synthesis of prostaglandins (PGs) from arachidonic acid [14], and prostaglandin E2 (PGE2) is chiefly responsible for the inflammatory response, which includes vasodilation, inflammatory cell

chemotaxis, and hyperalgesia [15]. Upregulated PGE2 also accelerates the COX pathway by positive feedback and leads to further COX2 production. Mucosal COX2 expression in irradiated mucosa is strongly correlated with the development of radiation-induced mucosal injury [16] and is considered an attractive target for repairing mucosal damage [17]. TJ-14 is a pharmaceutical formulation with ingredients active in the COX pathway, including *S. radix*, *Glycyrrhizae radix*, *C. rhizoma*, and *P. tuber* [18]. In particular, baicalin, baicalein, and wogonin, compounds found in *S. radix*, play primary roles in the inhibition of both COX2 activity and production of PGE2 [18, 19]. Glycyrrhizin in *Glycyrrhizae radix* [20] and ginsenosides in *Ginseng radix* [21, 22] also suppress COX activity by inducing corticosterone secretion. Therefore, the clinical benefit of TJ-14 against radiation-induced mucositis, in which COX2 activity was critical for its pathogenesis, was evident.

Robust infiltration of neutrophils in the irradiated mucosa was also suppressed in the TJ-14 treated group compared to that in the untreated group. During the ulcerative phase of mucositis, disruption of mucosal barriers causes the formation of a pseudomembrane where numerous microorganisms develop [13]. Among the active ingredients of TJ-14, berberine sulfate in *C. rhizoma* was

reported to exhibit wide-spectrum antimicrobial activity [23]. Infection and bacterial walls stimulate cytokine secretion and inflammatory cell chemotaxis, which further stimulate an intense inflammatory response. Therefore, our finding that TJ-14 inhibited neutrophil chemotaxis was thought to be due to the antimicrobial activity of TJ-14 along with its ability to inhibit COX2.

Altogether, TJ-14 exerted a preventative effect against radiation-induced mucositis by inhibiting the expression of COX2 and inflammatory cell chemotaxis via its anti-microbial activity. In addition, TJ-14 displayed an inhibitory effect on upregulated MAPK pathways in inflammatory tissue [24]. Therefore, the multi-targeted property of TJ-14 is an advantage for preventing mucositis.

In this study, we used a hamster model that has been established as an oral mucositis model for preclinical study [10, 25]. This animal model allows for selective irradiation to the buccal mucosa and easy accessibility for visual and histological evaluation. Moreover, by the habit of the animals, the oral diet was stored in the buccal pouch, which enabled sustained direct contact of TJ-14 with buccal mucosa. As Yamashita et al. emphasized [8], direct contact of TJ-14 with the mucosa was important for mucositis prevention and this model was suitable for simulating the clinical setting of TJ-14 application. Therefore, the results of our study reinforced the finding that TJ-14 had clinical benefit for radiation-induced mucositis as reported in a previous clinical study [8].

The number of clinical studies exploring the effectiveness of TJ-14 has recently increased, especially for cancer treatment-related mucositis, including chemotherapy-induced oral mucositis in colorectal cancer [9] and radiation-induced mucositis in HNC [8, 26]. The results of these studies indicated that TJ-14 attenuated mucositis, improved nutritional status, and increased the completion rate of cancer treatment. TJ-14 is advantageous because it is inexpensive, safe, easily applied, and efficacious for mucositis prevention. Therefore, TJ-14 is a novel and promising candidate for the prevention of oral mucositis.

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Compliance with ethical standards

Conflict of interest Authors declare no conflict of interest.

Ethical approval All procedures involving animals in this study were performed in accordance with the Guidelines for Proper Conduct of Animal Experiments and the US Public Health Service Policy on Human Care and Use of Laboratory Animals and were reviewed and approved by the Animal Ethical Committee of the National Defense Medical College (No. 13075).

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